

|         |                                                                 |                                                                                        |                                                                         |                                                                                                                                                 |  |  |  |  |  |
|---------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|
|         | Click here to track your progress : <a href="#">Course Link</a> |                                                                                        |                                                                         |                                                                                                                                                 |  |  |  |  |  |
|         |                                                                 |                                                                                        | Youtube Link                                                            | Articles                                                                                                                                        |  |  |  |  |  |
|         | 1. Basics of programming                                        |                                                                                        |                                                                         |                                                                                                                                                 |  |  |  |  |  |
|         | Problem Statement                                               | Leetcode / GFG Link                                                                    |                                                                         |                                                                                                                                                 |  |  |  |  |  |
|         | 1.1 Time and Space Complexity                                   |                                                                                        |                                                                         |                                                                                                                                                 |  |  |  |  |  |
| Pending | Learn about Time and Space Complexity                           | <a href="#">Theory [Watch the Video]</a>                                               | <a href="https://youtu.be/1mOwgPU-6A">https://youtu.be/1mOwgPU-6A</a>   | <a href="#">Time Complexity and Space Complexity in DSA   Explained with Examples - Code and Debug - Blog</a>                                   |  |  |  |  |  |
| Pending | What is TLE Error?                                              | <a href="#">Theory [Watch the Video]</a>                                               | <a href="https://youtu.be/VvAnuOzOSY">https://youtu.be/VvAnuOzOSY</a>   | -                                                                                                                                               |  |  |  |  |  |
| Pending | Time Complexity of Common Python Operations                     | <a href="#">Theory [Watch the Video]</a>                                               | <a href="https://youtu.be/PLV0f4RAMvI">https://youtu.be/PLV0f4RAMvI</a> | -                                                                                                                                               |  |  |  |  |  |
|         | 1.2 Basic Maths Logic Buildup                                   |                                                                                        |                                                                         |                                                                                                                                                 |  |  |  |  |  |
| Pending | Extraction of Digits                                            | <a href="#">Theory [Watch the Video]</a>                                               | <a href="https://youtu.be/yCMCdSzd1bY">https://youtu.be/yCMCdSzd1bY</a> | <a href="#">Extraction of Digits in Python   Explained Step-by-Step</a>                                                                         |  |  |  |  |  |
| Pending | Count Digits                                                    | <a href="#">Number of Digits - Naukri Code 360</a>                                     | <a href="https://youtu.be/ww-mxKUIltM">https://youtu.be/ww-mxKUIltM</a> | <a href="#">Python Program to Count Number of Digits   Explained</a>                                                                            |  |  |  |  |  |
| Pending | Reverse a number                                                | <a href="#">Reverse Integer - LeetCode</a>                                             |                                                                         | <a href="#">Leetcode #7 : Reverse Integer Python Program Explained</a>                                                                          |  |  |  |  |  |
| Pending | Check palindrome                                                | <a href="#">Palindrome Number - LeetCode</a>                                           | <a href="https://youtu.be/ecZVpAZ37UI">https://youtu.be/ecZVpAZ37UI</a> | <a href="#">Palindrome Number Program in Python   Leetcode #9</a>                                                                               |  |  |  |  |  |
| Pending | Armstrong Number                                                | <a href="#">Armstrong Number - LeetCode</a>                                            | <a href="https://youtu.be/rzRwie8UwQQ">https://youtu.be/rzRwie8UwQQ</a> | <a href="#">Python Program to Check Armstrong Number   Explained</a>                                                                            |  |  |  |  |  |
| Pending | Print all Divisors of a number                                  | <a href="#">Number of factors   Practice   GeeksforGeeks</a>                           | <a href="https://youtu.be/YoVIGYyWgyw">https://youtu.be/YoVIGYyWgyw</a> | <a href="#">Python Program to Print Divisors/Factors of an Integer</a>                                                                          |  |  |  |  |  |
| Pending | Check Prime                                                     | <a href="#">Prime Number   Practice   GeeksforGeeks</a>                                | <a href="#">(Do on your own)</a>                                        | <a href="#">Check if a Number is Prime or Not   GFG</a>                                                                                         |  |  |  |  |  |
|         | 1.3 Recursion Basics                                            |                                                                                        |                                                                         |                                                                                                                                                 |  |  |  |  |  |
| Pending | Introduction to Recursion                                       | <a href="#">Theory [Watch the Video]</a>                                               | <a href="https://youtu.be/-VwidRe3Ob8">https://youtu.be/-VwidRe3Ob8</a> | -                                                                                                                                               |  |  |  |  |  |
| Pending | Recursion using Parameters                                      | <a href="#">Theory [Watch the Video]</a>                                               | <a href="https://youtu.be/fgurDxhawRw">https://youtu.be/fgurDxhawRw</a> | -                                                                                                                                               |  |  |  |  |  |
| Pending | Functional Recursion                                            | <a href="#">Theory [Watch the Video]</a>                                               | <a href="https://youtu.be/UgksZgjPrD4">https://youtu.be/UgksZgjPrD4</a> | -                                                                                                                                               |  |  |  |  |  |
| Pending | Print 1 to N without Loop                                       | <a href="#">Print 1 To N Without Loop   Practice   GeeksforGeeks</a>                   |                                                                         | <a href="#">Python Program to Print from 1 to N Without Loops</a>                                                                               |  |  |  |  |  |
| Pending | Print N times with Recursion                                    | <a href="#">Print GFG n times   Practice   GeeksforGeeks</a>                           |                                                                         | <a href="#">Print N times with Recursion   GFG</a>                                                                                              |  |  |  |  |  |
| Pending | Print N to 1 without Loop                                       | <a href="#">Print N to 1 without loop   Practice   GeeksforGeeks</a>                   |                                                                         | <a href="#">Print N to 1 without loop   Normal Recursion and Backtracking</a>                                                                   |  |  |  |  |  |
| Pending | Sum of first N natural numbers without Loop                     | <a href="#">Sum of first n terms   Practice   GeeksforGeeks</a>                        |                                                                         | <a href="#">Sum of first n terms   Using Recursion   Explained</a>                                                                              |  |  |  |  |  |
| Pending | Factorial of N numbers                                          | <a href="#">Factorial   Practice   GeeksforGeeks</a>                                   | <a href="https://youtu.be/HKltpsIOOro">https://youtu.be/HKltpsIOOro</a> | <a href="#">Python Program to Find Factorial of Number Using Recursion</a>                                                                      |  |  |  |  |  |
| Pending | Reverse an Array without Loop                                   | <a href="#">Reverse sub array   Practice   GeeksforGeeks</a>                           | <a href="https://youtu.be/RynoW0e9I7U">https://youtu.be/RynoW0e9I7U</a> | <a href="#">Python Program to Reverse Array Using Recursion and While Loop</a>                                                                  |  |  |  |  |  |
| Pending | Check if String is palindrome                                   | <a href="#">Palindrome String   Practice   GeeksforGeeks</a>                           | <a href="https://youtu.be/uWYF8nLSE1A">https://youtu.be/uWYF8nLSE1A</a> | <a href="#">Palindrome String   Recursive &amp; Iterative Python Solutions</a>                                                                  |  |  |  |  |  |
| Pending | Fibonacci Series                                                | <a href="#">Fibonacci Number - LeetCode</a>                                            | <a href="https://youtu.be/33UUD-rkN9M">https://youtu.be/33UUD-rkN9M</a> | <a href="#">Fibonacci Number   Recursive Python Solution</a>                                                                                    |  |  |  |  |  |
|         | 1.4 Hashing                                                     |                                                                                        |                                                                         |                                                                                                                                                 |  |  |  |  |  |
| Pending | [Pre Requirement] How to store frequency                        | <a href="#">Theory [Watch the Video]</a>                                               | <a href="https://youtu.be/Ocd_uHpsQyg">https://youtu.be/Ocd_uHpsQyg</a> |                                                                                                                                                 |  |  |  |  |  |
| Pending | Learn basics of hashing                                         | <a href="#">Theory [Watch the Video]</a>                                               | <a href="https://youtu.be/0iEST63qEHg">https://youtu.be/0iEST63qEHg</a> |                                                                                                                                                 |  |  |  |  |  |
| Pending | Find the Frequency                                              | <a href="#">Find the Frequency   Practice   GeeksforGeeks</a>                          | <a href="#">(Do on your own)</a>                                        | -                                                                                                                                               |  |  |  |  |  |
| Pending | Count Frequency in a range                                      | <a href="#">Frequencies of Limited Range Array Elements   Practice   GeeksforGeeks</a> | <a href="#">(Do on your own)</a>                                        | -                                                                                                                                               |  |  |  |  |  |
|         | 2. Different types of Sorting                                   |                                                                                        |                                                                         |                                                                                                                                                 |  |  |  |  |  |
| Pending | Selection Sort                                                  | <a href="#">Selection Sort   Practice   GeeksforGeeks</a>                              | <a href="https://youtu.be/x8pZMsElvJk">https://youtu.be/x8pZMsElvJk</a> | <a href="#">Python Program for Selection Sort Algorithm</a>                                                                                     |  |  |  |  |  |
| Pending | Bubble Sort                                                     | <a href="#">Bubble Sort   Practice   GeeksforGeeks</a>                                 | <a href="https://youtu.be/M6Vg8SmGMA8">https://youtu.be/M6Vg8SmGMA8</a> | <a href="#">Bubble Sort Algorithm in Python   Explained in Depth</a>                                                                            |  |  |  |  |  |
| Pending | Insertion Sort                                                  | <a href="#">Insertion Sort   Practice   GeeksforGeeks</a>                              | <a href="https://youtu.be/C16LhvfOW">https://youtu.be/C16LhvfOW</a>     | <a href="#">Insertion Sort Algorithm   Explained in Python</a>                                                                                  |  |  |  |  |  |
| Pending | Merge Sort                                                      | <a href="#">Merge Sort   Practice   GeeksforGeeks</a>                                  | <a href="https://youtu.be/ZDCoxXeksWM">https://youtu.be/ZDCoxXeksWM</a> | <a href="#">Merge Sort Algorithm in Python   Explained</a>                                                                                      |  |  |  |  |  |
| Pending | Quick Sort                                                      | <a href="#">Quick Sort   Practice   GeeksforGeeks</a>                                  | <a href="https://youtu.be/77tO3JkVeS0">https://youtu.be/77tO3JkVeS0</a> | <a href="#">Quick Sort Algorithm in Python   Explained</a>                                                                                      |  |  |  |  |  |
| Pending | Recursive Bubble Sort                                           | <a href="#">Bubble Sort   Practice   GeeksforGeeks</a>                                 | <a href="#">(Do on your own)</a>                                        | <a href="#">Recursive Bubble Sort in Python</a>                                                                                                 |  |  |  |  |  |
| Pending | Recursive Insertion Sort                                        | <a href="#">Insertion Sort   Practice   GeeksforGeeks</a>                              | <a href="#">(Do on your own)</a>                                        | <a href="#">Recursive Insertion Sort in Python</a>                                                                                              |  |  |  |  |  |
|         | 3. Problems on Arrays                                           |                                                                                        |                                                                         |                                                                                                                                                 |  |  |  |  |  |
|         | 3.1 EASY                                                        |                                                                                        |                                                                         |                                                                                                                                                 |  |  |  |  |  |
| Pending | Largest Element in an Array                                     | <a href="#">Largest Element in Array   Practice   GeeksforGeeks</a>                    | <a href="https://youtu.be/hS2p90uVUGA">https://youtu.be/hS2p90uVUGA</a> | <a href="#">Find the Largest Element in an Array (Brute Force &amp; Optimal Solutions)   Step-by-Step with Images</a>                           |  |  |  |  |  |
| Pending | Second Largest Element in an Array without sorting              | <a href="#">Second Largest Number - Naukri Code 360</a>                                | <a href="https://youtu.be/Z6TWNJ9OoQ">https://youtu.be/Z6TWNJ9OoQ</a>   | <a href="#">Find the second largest and second smallest element in an Array   Explained</a>                                                     |  |  |  |  |  |
| Pending | Check if the array is sorted                                    | <a href="#">Check if array is sorted   Practice   GeeksforGeeks</a>                    | <a href="https://youtu.be/_T4t72Y6yd4">https://youtu.be/_T4t72Y6yd4</a> | <a href="#">Check if array is sorted   Explained using Python Code</a>                                                                          |  |  |  |  |  |
| Pending | Remove duplicates from Sorted array                             | <a href="#">Remove Duplicates from Sorted Array - LeetCode</a>                         | <a href="https://youtu.be/a2gP9uYrpQo">https://youtu.be/a2gP9uYrpQo</a> | <a href="#">Remove duplicates from Sorted array - Leetcode 26</a>                                                                               |  |  |  |  |  |
| Pending | Right Rotate an array by one place                              | <a href="#">Theory [Watch the Video]</a>                                               | <a href="https://youtu.be/_EHlC6xhhE0">https://youtu.be/_EHlC6xhhE0</a> | -                                                                                                                                               |  |  |  |  |  |
| Pending | Right rotate an array by K places                               | <a href="#">Rotate Array - LeetCode</a>                                                | <a href="https://youtu.be/HRzQK5trZM">https://youtu.be/HRzQK5trZM</a>   | <a href="#">Right Rotate an Array by K places   Explained with Examples</a>                                                                     |  |  |  |  |  |
| Pending | Move Zeros to end                                               | <a href="#">Move Zeroes - LeetCode</a>                                                 | <a href="https://youtu.be/GtssSuFh_Ek">https://youtu.be/GtssSuFh_Ek</a> | <a href="#">Move Zeroes   Leetcode 283   Explained</a>                                                                                          |  |  |  |  |  |
| Pending | Linear Search                                                   | <a href="#">Array Search   Practice   GeeksforGeeks</a>                                | <a href="https://youtu.be/mjijlUK0ZIE">https://youtu.be/mjijlUK0ZIE</a> | <a href="#">Linear Search   Explained in Python</a>                                                                                             |  |  |  |  |  |
| Pending | Merge 2 sorted Arrays                                           | <a href="#">Union of Two Sorted Arrays   Practice   GeeksforGeeks</a>                  | <a href="https://youtu.be/t9RpK0j9ofs">https://youtu.be/t9RpK0j9ofs</a> | <a href="#">Union of 2 Sorted with Duplicates   Explained with Images</a>                                                                       |  |  |  |  |  |
| Pending | Find missing number in an array                                 | <a href="#">Missing Number - LeetCode</a>                                              | <a href="https://youtu.be/Zold2EqX45U">https://youtu.be/Zold2EqX45U</a> | <a href="#">Missing Number   Leetcode 268   Explained</a>                                                                                       |  |  |  |  |  |
| Pending | Maximum Consecutive Ones                                        | <a href="#">Max Consecutive Ones - LeetCode</a>                                        | <a href="https://youtu.be/hQLnjtdEi9Y">https://youtu.be/hQLnjtdEi9Y</a> | <a href="#">Maximum Consecutive Ones   Leetcode 485   Explained with Images</a>                                                                 |  |  |  |  |  |
|         | 3.2 MEDIUM                                                      |                                                                                        |                                                                         |                                                                                                                                                 |  |  |  |  |  |
| Pending | 2Sum Problem                                                    | <a href="#">Two Sum - LeetCode</a>                                                     | <a href="https://youtu.be/EX2tR_cgCz4">https://youtu.be/EX2tR_cgCz4</a> | <a href="https://codeanddebug.in/blog/two-sum-leetcode-1/">https://codeanddebug.in/blog/two-sum-leetcode-1/</a>                                 |  |  |  |  |  |
| Pending | Kadane's Algorithm, maximum subarray sum                        | <a href="#">Maximum Subarray - LeetCode</a>                                            | <a href="https://youtu.be/63luQx08W8">https://youtu.be/63luQx08W8</a>   | <a href="https://codeanddebug.in/blog/maximum-subarray-kadanes-algorithm/">https://codeanddebug.in/blog/maximum-subarray-kadanes-algorithm/</a> |  |  |  |  |  |
| Pending | Stock Buy and Sell                                              | <a href="#">Best Time to Buy and Sell Stock - LeetCode</a>                             | <a href="https://youtu.be/tGrDqGxWsmc">https://youtu.be/tGrDqGxWsmc</a> | <a href="#">Best Time to Buy and Sell Stock   Leetcode 121   Explained with Images</a>                                                          |  |  |  |  |  |
| Pending | Rearrange the array in alternating positive and negative items  | <a href="#">Rearrange Array Elements by Sign - LeetCode</a>                            | <a href="https://youtu.be/xFmITgc-HTI">https://youtu.be/xFmITgc-HTI</a> | <a href="#">Rearrange Array Elements by Sign   Leetcode 2149</a>                                                                                |  |  |  |  |  |
| Pending | Longest Consecutive Sequence in an Array                        | <a href="#">Longest Consecutive Sequence - LeetCode</a>                                | <a href="https://youtu.be/OO2FsBeFcEs">https://youtu.be/OO2FsBeFcEs</a> | <a href="#">Longest Consecutive Sequence   Leetcode 128</a>                                                                                     |  |  |  |  |  |
| Pending | Learn 2D Matrix [Check video]                                   | <a href="#">Theory [Watch the Video]</a>                                               | <a href="https://youtu.be/sVzamPflrw">https://youtu.be/sVzamPflrw</a>   |                                                                                                                                                 |  |  |  |  |  |
| Pending | Set Matrix Zeros                                                | <a href="#">Set Matrix Zeroes - LeetCode</a>                                           | <a href="https://youtu.be/_RMrPINBrA">https://youtu.be/_RMrPINBrA</a>   | <a href="#">Set Matrix Zeroes   Leetcode 73   Explained</a>                                                                                     |  |  |  |  |  |
| Pending | Rotate Matrix by 90 degrees                                     | <a href="#">Rotate Image - LeetCode</a>                                                | <a href="https://youtu.be/w9oZwey57M">https://youtu.be/w9oZwey57M</a>   | <a href="#">Rotate Image   Leetcode 48   Explained with Images</a>                                                                              |  |  |  |  |  |
| Pending | Print the matrix in spiral manner                               | <a href="#">Spiral Matrix - LeetCode</a>                                               | <a href="https://youtu.be/FHnZAvZN-Co">https://youtu.be/FHnZAvZN-Co</a> | <a href="#">Spiral Matrix   Leetcode 54   Explained in Python</a>                                                                               |  |  |  |  |  |







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| Pending                                     | Word ladder - 2                                                             | Word Ladder II - LeetCode                                                              | <a href="https://youtu.be/tkPYpeLI4_Q">https://youtu.be/tkPYpeLI4_Q</a>   | Word Ladder II   Leetcode 126   Shortest Transformation Sequences with BFS                                                                                                                                                        |  |  |  |  |
| Pending                                     | Number of Islands                                                           | Number of Islands - LeetCode                                                           | <a href="https://youtu.be/vh_eWEIEI8">https://youtu.be/vh_eWEIEI8</a>     | Number of Islands   Leetcode 200   Explained using BFS and DFS                                                                                                                                                                    |  |  |  |  |
| Pending                                     | Number of Distinct Islands [dfs multisource]                                | Number of Distinct Islands   Practice   GeeksforGeeks                                  | <a href="https://youtu.be/mMmI3BAflc">https://youtu.be/mMmI3BAflc</a>     | Number of Distinct Islands   Simple DFS solution in Python                                                                                                                                                                        |  |  |  |  |
| Pending                                     | Bipartite Graph (DFS)                                                       | Is Graph Bipartite? - LeetCode                                                         | <a href="https://youtu.be/tJxPMjy5eEM">https://youtu.be/tJxPMjy5eEM</a>   | Is Graph Bipartite?   Leetcode 785   Easy 2-Color DFS Solution in Python                                                                                                                                                          |  |  |  |  |
| Pending                                     | Cycle Detection in Directed Graph (DFS)                                     | Directed Graph Cycle   Practice   GeeksforGeeks                                        | <a href="https://youtu.be/BrxY15Gz2DY">https://youtu.be/BrxY15Gz2DY</a>   | Detect Cycle in a Directed Graph                                                                                                                                                                                                  |  |  |  |  |
| 15.3 Topo Sort based problems               |                                                                             |                                                                                        |                                                                           |                                                                                                                                                                                                                                   |  |  |  |  |
| Pending                                     | Topo Sort                                                                   | Topological sort   Practice   GeeksforGeeks                                            | <a href="https://youtu.be/hy7IViJNkzg">https://youtu.be/hy7IViJNkzg</a>   | Topological Sort in Graph   Simple DFS + Stack Solution in Python                                                                                                                                                                 |  |  |  |  |
| Pending                                     | Kahn's Algorithm                                                            | Topological sort   Practice   GeeksforGeeks                                            | <a href="https://youtu.be/5E89BR5Q-o">https://youtu.be/5E89BR5Q-o</a>     | Topological Sort (Kahn's Algorithm) - Simple BFS Solution in Python                                                                                                                                                               |  |  |  |  |
| Pending                                     | Cycle Detection in Directed Graph (BFS)                                     | Detect cycle in a directed graph   Practice   GeeksforGeeks                            | <a href="https://youtu.be/ZrwQaFY_4rE">https://youtu.be/ZrwQaFY_4rE</a>   | Detect Cycle in a Directed Graph - Kahn's Algorithm (BFS Topological Check) in Python                                                                                                                                             |  |  |  |  |
| Pending                                     | Course Schedule - I                                                         | Course Schedule - LeetCode                                                             | <a href="https://youtu.be/q2mauGGPmwl">https://youtu.be/q2mauGGPmwl</a>   | Course Schedule   Leetcode 207   Kahn's Algorithm Walk-through in Python                                                                                                                                                          |  |  |  |  |
| Pending                                     | Course Schedule - II                                                        | Course Schedule II - LeetCode                                                          | <a href="https://youtu.be/q2mauGGPmwl">https://youtu.be/q2mauGGPmwl</a>   |                                                                                                                                                                                                                                   |  |  |  |  |
| Pending                                     | Find eventual safe states (BFS)                                             | Find Eventual Safe States - LeetCode                                                   | <a href="https://youtu.be/b3LIEOPH4Nc">https://youtu.be/b3LIEOPH4Nc</a>   | Find Eventual Safe States   Leetcode 802   Kahn's Algorithm (BFS) Solution                                                                                                                                                        |  |  |  |  |
| Pending                                     | Find eventual safe states (DFS)                                             | Find Eventual Safe States - LeetCode                                                   | <a href="https://youtu.be/-MIOLKzruRY">https://youtu.be/-MIOLKzruRY</a>   | Find Eventual Safe States   Leetcode 802   DFS Solution                                                                                                                                                                           |  |  |  |  |
| Pending                                     | Alien dictionary                                                            | Alien Dictionary   Practice   GeeksforGeeks                                            | <a href="https://youtu.be/vJRRRT13Bmo">https://youtu.be/vJRRRT13Bmo</a>   | [Do on your own]                                                                                                                                                                                                                  |  |  |  |  |
| 15.4 Shortest Path related Problems         |                                                                             |                                                                                        |                                                                           |                                                                                                                                                                                                                                   |  |  |  |  |
| Pending                                     | Shortest Path in UG with unit weights                                       | Shortest path in Undirected Graph   Practice   GeeksforGeeks                           | <a href="https://youtu.be/dtR5EdDT7do0">https://youtu.be/dtR5EdDT7do0</a> | Shortest Path in an Undirected Graph with Unit Distance - Clear BFS Solution in Python                                                                                                                                            |  |  |  |  |
| Pending                                     | Shortest Path in DAG                                                        | Shortest path in Directed Acyclic Graph   Practice   GeeksforGeeks                     | <a href="https://youtu.be/b8IB5UJp4U">https://youtu.be/b8IB5UJp4U</a>     | Shortest Path in a Weighted Directed Acyclic Graph   Topological-Sort + Relaxation   Python                                                                                                                                       |  |  |  |  |
| Pending                                     | Dijkstra's Algorithm (Using Priority Queue)                                 | Implementing Dijkstra Algorithm   Practice   GeeksforGeeks                             | <a href="https://youtu.be/0Xr7BFg22k">https://youtu.be/0Xr7BFg22k</a>     | Dijkstra's Algorithm with a Priority Queue                                                                                                                                                                                        |  |  |  |  |
| Pending                                     | Dijkstra's Algorithm (Using Set)                                            | Implementing Dijkstra Algorithm   Practice   GeeksforGeeks                             | <a href="https://youtu.be/2BLYnNypMA">https://youtu.be/2BLYnNypMA</a>     | Dijkstra's Algorithm in Python Using a Set                                                                                                                                                                                        |  |  |  |  |
| Pending                                     | Why priority Queue is used in Dijkstra's Algorithm                          | Implementing Dijkstra Algorithm   Practice   GeeksforGeeks                             | <a href="https://youtu.be/kb4O-6cbWZg">https://youtu.be/kb4O-6cbWZg</a>   | [Watch Video]                                                                                                                                                                                                                     |  |  |  |  |
| Pending                                     | Print the Shortest Path                                                     | Shortest Path in Weighted undirected graph   Practice   GeeksforGeeks                  | <a href="https://youtu.be/laRgnK9uwCE">https://youtu.be/laRgnK9uwCE</a>   | Printing the Actual Shortest Path with Dijkstra Algorithm                                                                                                                                                                         |  |  |  |  |
| Pending                                     | Shortest path in a binary maze                                              | Shortest Path in Binary Matrix - LeetCode                                              | <a href="https://youtu.be/u6lPL25Q0Fo">https://youtu.be/u6lPL25Q0Fo</a>   | Shortest Path in a Binary Matrix   Leetcode 1091   Dijkstra's Algorithm with Normal Queue (BFS)                                                                                                                                   |  |  |  |  |
| Pending                                     | Path with minimum effort                                                    | Path With Minimum Effort - LeetCode                                                    | <a href="https://youtu.be/scW82V_eFY">https://youtu.be/scW82V_eFY</a>     | Path With Minimum Effort   Leetcode 1631   Explained                                                                                                                                                                              |  |  |  |  |
| Pending                                     | Cheapest flights within k stops                                             | Cheapest Flights Within K Stops - LeetCode                                             | <a href="https://youtu.be/8UJ4iq3LBUw">https://youtu.be/8UJ4iq3LBUw</a>   | Cheapest Flights Within K Stops   Leetcode 787   Easy BFS Approach in Python - Code and Debug - Blog                                                                                                                              |  |  |  |  |
| Pending                                     | Minimum steps to reach end from start by performing multiple jumps          | Minimum Multiplications to reach End   Practice   GeeksforGeeks                        | <a href="https://youtu.be/H5p2KqvQ6l">https://youtu.be/H5p2KqvQ6l</a>     | <a href="https://codeanddebug.in/blog/minimum-multiplications-to-reach-end-python-bfs/">https://codeanddebug.in/blog/minimum-multiplications-to-reach-end-python-bfs/</a>                                                         |  |  |  |  |
| Pending                                     | Number of ways to arrive at destination                                     | Number of Ways to Arrive at Destination - LeetCode                                     | <a href="https://youtu.be/7vFT5Y4uGM">https://youtu.be/7vFT5Y4uGM</a>     | <a href="https://codeanddebug.in/blog/number-of-ways-to-arrive-at-destination-python/">https://codeanddebug.in/blog/number-of-ways-to-arrive-at-destination-python/</a>                                                           |  |  |  |  |
| Pending                                     | Bellman Ford Algorithm                                                      | Distance from the Source (Bellman-Ford Algorithm)   Practice   GeeksforGeeks           | <a href="https://youtu.be/7C_wSRiEyoU">https://youtu.be/7C_wSRiEyoU</a>   | <a href="https://codeanddebug.in/blog/bellman-ford-algorithm/">https://codeanddebug.in/blog/bellman-ford-algorithm/</a>                                                                                                           |  |  |  |  |
| Pending                                     | Floyd Warshal Algorithm                                                     | Floyd Warshall   Practice   GeeksforGeeks                                              | <a href="https://youtu.be/LULxlvOhGc">https://youtu.be/LULxlvOhGc</a>     | <a href="https://codeanddebug.in/blog/floyd-warshall-vs-dijkstra-all-pairs-shortest-paths/">https://codeanddebug.in/blog/floyd-warshall-vs-dijkstra-all-pairs-shortest-paths/</a>                                                 |  |  |  |  |
| Pending                                     | Find the city with the smallest number of neighbors in a threshold distance | Find the City With the Smallest Number of Neighbors at a Threshold Distance - LeetCode | <a href="https://youtu.be/LkudkoXzOXY">https://youtu.be/LkudkoXzOXY</a>   | <a href="https://codeanddebug.in/blog/find-the-city-with-the-smallest-number-of-neighbors-at-a-threshold-distance/">https://codeanddebug.in/blog/find-the-city-with-the-smallest-number-of-neighbors-at-a-threshold-distance/</a> |  |  |  |  |
| 15.5 Minimum Spanning Tree and Disjoint Set |                                                                             |                                                                                        |                                                                           |                                                                                                                                                                                                                                   |  |  |  |  |
| Pending                                     | Minimum Spanning Tree                                                       | Minimum Spanning Tree   Practice   GeeksforGeeks                                       | <a href="https://youtu.be/Q-FY2yCRhvk">https://youtu.be/Q-FY2yCRhvk</a>   |                                                                                                                                                                                                                                   |  |  |  |  |
| Pending                                     | Prim's Algorithm                                                            | Minimum Spanning Tree   Practice   GeeksforGeeks                                       | <a href="https://youtu.be/t2CNZcsYR7o">https://youtu.be/t2CNZcsYR7o</a>   |                                                                                                                                                                                                                                   |  |  |  |  |
| Pending                                     | Disjoint Set [Union by Rank]                                                | Disjoint set (Union-Find)   Practice   GeeksforGeeks                                   | <a href="https://youtu.be/E...JlsIBTik">https://youtu.be/E...JlsIBTik</a> |                                                                                                                                                                                                                                   |  |  |  |  |
| Pending                                     | Disjoint Set [Union by Size]                                                | Disjoint set (Union-Find)   Practice   GeeksforGeeks                                   | <a href="https://youtu.be/E...JlsIBTik">https://youtu.be/E...JlsIBTik</a> |                                                                                                                                                                                                                                   |  |  |  |  |
| Pending                                     | Kruskal's Algorithm                                                         | Minimum Spanning Tree   Practice   GeeksforGeeks                                       | <a href="https://youtu.be/LdknY_dftyo">https://youtu.be/LdknY_dftyo</a>   |                                                                                                                                                                                                                                   |  |  |  |  |
| Pending                                     | Number of Provinces                                                         | Number of Provinces   Practice   GeeksforGeeks                                         | <a href="https://youtu.be/5OmFFCooWlw">https://youtu.be/5OmFFCooWlw</a>   |                                                                                                                                                                                                                                   |  |  |  |  |
| Pending                                     | Number of operations to make network connected                              | Number of Operations to Make Network Connected - LeetCode                              | <a href="https://youtu.be/RiW-WvWwO3k">https://youtu.be/RiW-WvWwO3k</a>   |                                                                                                                                                                                                                                   |  |  |  |  |
| 16. Dynamic Programming                     |                                                                             |                                                                                        |                                                                           |                                                                                                                                                                                                                                   |  |  |  |  |
| 16.1 Introduction to Dynamic Programming    |                                                                             |                                                                                        |                                                                           |                                                                                                                                                                                                                                   |  |  |  |  |
| Pending                                     | Dynamic Programming Introduction                                            | Nth Fibonacci Number   Practice   GeeksforGeeks                                        | <a href="https://youtu.be/-CK7HfwqhLS">https://youtu.be/-CK7HfwqhLS</a>   | Nth Fibonacci Number using Dynamic Programming                                                                                                                                                                                    |  |  |  |  |
| 16.2 DP on 1D                               |                                                                             |                                                                                        |                                                                           |                                                                                                                                                                                                                                   |  |  |  |  |
| Pending                                     | Climbing Stars                                                              | Climbing Stairs - LeetCode                                                             | <a href="https://youtu.be/vB_Q4VWYRg">https://youtu.be/vB_Q4VWYRg</a>     | Climbing Stairs   Leetcode 70   Solved using Dynamic Programming                                                                                                                                                                  |  |  |  |  |
| Pending                                     | Frog Jump                                                                   | Geek Jump   Practice   GeeksforGeeks                                                   | <a href="https://youtu.be/ozl5yIjeJRg">https://youtu.be/ozl5yIjeJRg</a>   | <a href="https://codeanddebug.in/blog/frog-jump/">https://codeanddebug.in/blog/frog-jump/</a>                                                                                                                                     |  |  |  |  |
| Pending                                     | Maximum sum of non-adjacent elements                                        | House Robber - LeetCode                                                                | <a href="https://youtu.be/eyRPBmmUAc4">https://youtu.be/eyRPBmmUAc4</a>   | <a href="https://codeanddebug.in/blog/house-robber/">https://codeanddebug.in/blog/house-robber/</a>                                                                                                                               |  |  |  |  |
| Pending                                     | House Robber 2                                                              | House Robber II - LeetCode                                                             | <a href="https://youtu.be/9DXrm35Cvds">https://youtu.be/9DXrm35Cvds</a>   | House Robber II   Leetcode 213   All 4 Solutions                                                                                                                                                                                  |  |  |  |  |
| 16.3 DP on 2D/3D or Grids                   |                                                                             |                                                                                        |                                                                           |                                                                                                                                                                                                                                   |  |  |  |  |
| Pending                                     | Ninja's Training                                                            | Geek's Training   Practice   GeeksforGeeks                                             | <a href="https://youtu.be/fbiWcpuzvYs">https://youtu.be/fbiWcpuzvYs</a>   | Geek's Training   2D Dynamic Programming   GFG Practice                                                                                                                                                                           |  |  |  |  |
| Pending                                     | Grid Unique Paths : DP on Grids                                             | Unique Paths - LeetCode                                                                | <a href="https://youtu.be/dZNV9dDvo4g">https://youtu.be/dZNV9dDvo4g</a>   | Unique Paths   Leetcode 62   All 4 DP Solutions                                                                                                                                                                                   |  |  |  |  |
| Pending                                     | Grid Unique Paths 2                                                         | Unique Paths II - LeetCode                                                             | <a href="https://youtu.be/DBqyEdYTLI">https://youtu.be/DBqyEdYTLI</a>     | Unique Paths II   Leetcode 63   All 4 DP Solutions                                                                                                                                                                                |  |  |  |  |
| Pending                                     | Minimum path sum in Grid                                                    | Minimum Path Sum - LeetCode                                                            | <a href="https://youtu.be/bLcFWUt2gFY">https://youtu.be/bLcFWUt2gFY</a>   | Minimum Path Sum   Leetcode 64   All 4 DP Solutions                                                                                                                                                                               |  |  |  |  |
| Pending                                     | Minimum path sum in Triangular Grid                                         | Triangle - LeetCode                                                                    | <a href="https://youtu.be/mEbfHn5QA">https://youtu.be/mEbfHn5QA</a>       | Triangle   Leetcode 120   All 4 DP Solutions                                                                                                                                                                                      |  |  |  |  |
| Pending                                     | Minimum/Maximum Falling Path Sum                                            | Minimum Falling Path Sum - LeetCode                                                    | <a href="https://youtu.be/cB8jXNcN9Y">https://youtu.be/cB8jXNcN9Y</a>     | Minimum Falling Path Sum   Leetcode 931   Variable Starting and Ending Points                                                                                                                                                     |  |  |  |  |
| Pending                                     | 3-d DP : Ninja and his friends                                              | Cherry Pickup II - LeetCode                                                            | <a href="https://youtu.be/R2wbmFgbQMY">https://youtu.be/R2wbmFgbQMY</a>   | Cherry Pickup II   Leetcode 1463   Solved using Dynamic Programming                                                                                                                                                               |  |  |  |  |
| 16.4 DP on subsequences                     |                                                                             |                                                                                        |                                                                           |                                                                                                                                                                                                                                   |  |  |  |  |
| Pending                                     | Subset sum equal to target                                                  | Subset Sum Problem   Practice   GeeksforGeeks                                          | <a href="https://youtu.be/GDG84KiUfUQ">https://youtu.be/GDG84KiUfUQ</a>   | Subset Sum Problem   Solved using Dynamic Programming                                                                                                                                                                             |  |  |  |  |
| Pending                                     | Partition Equal Subset Sum                                                  | Partition Equal Subset Sum - LeetCode                                                  | <a href="https://youtu.be/s7L6SSYkHc">https://youtu.be/s7L6SSYkHc</a>     | Partition Equal Subset Sum   Leetcode 416                                                                                                                                                                                         |  |  |  |  |
| Pending                                     | Partition Set Into 2 Subsets With Min Absolute Sum Diff                     | Minimum sum partition   Practice   GeeksforGeeks                                       | <a href="https://youtu.be/hc_PikbISIM">https://youtu.be/hc_PikbISIM</a>   | Minimum sum partition   Solved using Tabulation in Python                                                                                                                                                                         |  |  |  |  |
| Pending                                     | Count Subsets with Sum K                                                    | Perfect Sum Problem   Practice   GeeksforGeeks                                         | <a href="https://youtu.be/-OdBMAZ54HY">https://youtu.be/-OdBMAZ54HY</a>   | Perfect Sum Problem   All 4 DP Solutions in Python                                                                                                                                                                                |  |  |  |  |
| Pending                                     | Count Partitions with Given Difference                                      | Partitions with Given Difference   Practice   GeeksforGeeks                            | <a href="https://youtu.be/4EaT73nr62w">https://youtu.be/4EaT73nr62w</a>   | Partitions with Given Difference   Optimal Solution                                                                                                                                                                               |  |  |  |  |
| Pending                                     | 0/1 Knapsack                                                                | 0 - 1 Knapsack Problem   Practice   GeeksforGeeks                                      | <a href="https://youtu.be/pqLiITVUYes">https://youtu.be/pqLiITVUYes</a>   | 0 - 1 Knapsack Problem   5 DP Solutions in Python                                                                                                                                                                                 |  |  |  |  |
| 16.5 DP on Strings                          |                                                                             |                                                                                        |                                                                           |                                                                                                                                                                                                                                   |  |  |  |  |
| Pending                                     | Longest Common Subsequence                                                  | Longest Common Subsequence - LeetCode                                                  | <a href="https://youtu.be/GLIFaLo73c">https://youtu.be/GLIFaLo73c</a>     | Longest Common Subsequence   Leetcode 1143   All DP Approach                                                                                                                                                                      |  |  |  |  |
| Pending                                     | Print Longest Common Subsequence                                            | Print all LCS sequences   Practice   GeeksforGeeks                                     | <a href="https://youtu.be/QvOZxe2L90I">https://youtu.be/QvOZxe2L90I</a>   | Print the Longest Common Subsequence   Build Table, Then Reconstruct                                                                                                                                                              |  |  |  |  |

|         |                                                |                                                                                |                                                                           |                                                                                                      |  |  |  |  |
|---------|------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--|--|--|--|
| Pending | Longest Common Substring                       | <a href="#">Longest Common Substring   Practice   GeeksforGeeks</a>            | <a href="https://youtu.be/4sArpzW0txs">https://youtu.be/4sArpzW0txs</a>   | <a href="#">Longest Common Substring   Solved using Tabulation</a>                                   |  |  |  |  |
| Pending | Longest Palindromic Subsequence                | <a href="#">Longest Palindromic Subsequence - LeetCode</a>                     | <a href="https://youtu.be/RXcASvYIPX8">https://youtu.be/RXcASvYIPX8</a>   | <a href="#">Longest Palindromic Subsequence   Leetcode 516</a>                                       |  |  |  |  |
| Pending | Minimum insertions to make string palindrome   | <a href="#">Minimum Insertion Steps to Make a String Palindrome - LeetCode</a> | <a href="https://youtu.be/790c950eDMl">https://youtu.be/790c950eDMl</a>   | <a href="#">Minimum Insertion Steps to Make a String Palindrome   Leetcode 1312</a>                  |  |  |  |  |
| Pending | Minimum Insertions/Deletions to Convert String | <a href="#">Delete Operation for Two Strings - LeetCode</a>                    | <a href="https://youtu.be/ZHUeIQR2eK8">https://youtu.be/ZHUeIQR2eK8</a>   | <a href="#">Delete Operation for Two Strings   Leetcode 583</a>                                      |  |  |  |  |
|         |                                                |                                                                                |                                                                           |                                                                                                      |  |  |  |  |
|         |                                                | <b>16.6 DP on Stocks</b>                                                       |                                                                           |                                                                                                      |  |  |  |  |
|         |                                                | <a href="#">Best Time to Buy and Sell Stock - LeetCode</a>                     |                                                                           |                                                                                                      |  |  |  |  |
| Pending | Best Time to Buy and Sell Stock                | <a href="#">Best Time to Buy and Sell Stock II - LeetCode</a>                  | <a href="#">Already Done in Arrays Section</a>                            | -                                                                                                    |  |  |  |  |
| Pending | Buy and Sell Stock - II                        | <a href="#">Best Time to Buy and Sell Stock III - LeetCode</a>                 | <a href="https://youtu.be/QE6tM6NXMRw">https://youtu.be/QE6tM6NXMRw</a>   | <a href="#">Best Time to Buy and Sell Stock II   Leetcode 122   Recursion to Tabulation</a>          |  |  |  |  |
| Pending | Buy and Sell Stocks III                        | <a href="#">Best Time to Buy and Sell Stock IV - LeetCode</a>                  | <a href="https://youtu.be/hKhfbZ0zlaQw">https://youtu.be/hKhfbZ0zlaQw</a> | <a href="#">Best Time to Buy and Sell Stock III   Leetcode 123   Dynamic Programming Solution</a>    |  |  |  |  |
| Pending | Buy and Stock Sell IV                          |                                                                                | <a href="https://youtu.be/7BSuRfRNGzw">https://youtu.be/7BSuRfRNGzw</a>   | <a href="#">Best Time to Buy and Sell Stock IV   Leetcode 188   Recursion to Tabulation Approach</a> |  |  |  |  |
|         |                                                |                                                                                |                                                                           |                                                                                                      |  |  |  |  |
|         |                                                | <b>17. Tries</b>                                                               |                                                                           |                                                                                                      |  |  |  |  |
|         |                                                | <b>17.1 Introduction and Problems</b>                                          |                                                                           |                                                                                                      |  |  |  |  |
| Pending | Implement TRIE   INSERT   SEARCH   STARTSWITH  | <a href="#">Implement Trie (Prefix Tree) - LeetCode</a>                        | <a href="https://youtu.be/qKFGfDLE3mA">https://youtu.be/qKFGfDLE3mA</a>   | <a href="#">Implement Trie (Prefix Tree)   Leetcode 208   Efficient String Data Structure</a>        |  |  |  |  |
| Pending | Implement Trie - 2 (Prefix Tree)               | <a href="#">Implement Trie II - Naukri Code 360</a>                            | <a href="https://youtu.be/9gDAwZz2nZ4">https://youtu.be/9gDAwZz2nZ4</a>   | <a href="#">Implement Trie II   Full Trie with Insert, Count, and Erase</a>                          |  |  |  |  |
| Pending | Longest String with All Prefixes               | <a href="#">Complete String - Naukri Code 360</a>                              | <a href="https://youtu.be/hTzGe6s0SU">https://youtu.be/hTzGe6s0SU</a>     | <a href="#">Complete String - Trie Based Solution</a>                                                |  |  |  |  |
| Pending | Number of Distinct Substrings in a String      | <a href="#">Count Distinct Substrings - Naukri Code 360</a>                    | <a href="https://youtu.be/f9PkrK0pQoU">https://youtu.be/f9PkrK0pQoU</a>   | <a href="#">Count Distinct Substrings - Brute Force and Trie Approaches</a>                          |  |  |  |  |